

UNIT V: Association Rule Mining

Frequent pattern mining is a key data mining technique used to discover **frequent patterns, associations, correlations, or causal structures** among sets of items in transactional or relational databases. Its primary goal is to find itemsets that appear **frequently together** in a dataset.

Term	Explanation
Frequent Itemsets	Sets of items that appear together in transactions with frequency (support) above a threshold.
Closed Itemsets	Frequent itemsets where none of its supersets has the same support count. They represent maximal information without redundancy.
Association Rules	Implication expressions of the form $A \Rightarrow B$, meaning if itemset A occurs, then itemset B is likely to occur as well.

Example: Consider a transaction dataset:

Transaction ID	Items Bought
1	Bread, Milk
2	Bread, Diaper, Beer
3	Milk, Diaper, Beer
4	Bread, Milk, Diaper
5	Bread, Milk, Diaper, Beer

- **Frequent Itemset:** {Bread, Milk} appears 3 times (support = $3/5 = 60\%$)
- **Closed Itemset:** {Bread, Milk, Diaper} appears 2 times (support = $2/5 = 40\%$) and none of its supersets have same support.
- **Association Rule:** {Diaper} $\Rightarrow$ {Beer} with support = $3/5 = 60\%$ and confidence = $\text{support}(\{\text{Diaper, Beer}\}) / \text{support}(\{\text{Diaper}\}) = 3/4 = 75\%$.

Apriori Algorithm	Eclat Algorithm
Uses a breadth-first search to generate candidate itemsets	Uses a depth-first search to explore itemsets recursively
Works on horizontal data format (transaction list)	Works on vertical data format (tid-lists or transaction IDs)
Generates candidate itemsets explicitly	Uses intersections of tid-lists to find frequent itemsets
Generally higher memory usage due to candidate generation	More memory efficient due to tid-lists and recursion
Slower due to multiple scans of database and candidate generation	Faster for dense datasets due to efficient intersection operations
High computational cost in candidate generation and pruning	More efficient in handling large datasets
Easier to implement	complex due to recursive process

FP-Growth is **frequent pattern mining** algorithm that **efficiently discovers frequent itemsets** without candidate generation. It uses a **compact data structure called FP-Tree (Frequent Pattern Tree)** to represent the database, and mines the frequent patterns directly from this tree.

How FP-Growth Works:

1. **Build FP-Tree:** Scan the database once to find frequent items (above minimum support). Sort frequent items in descending order of support. Create an FP-Tree by inserting transactions, sharing common prefixes.
2. **Mine FP-Tree:** Extract frequent itemsets by recursively exploring conditional FP-Trees (subtrees conditioned on a particular item).

FP-Growth	Apriori
No candidate generation	Generates large numbers of candidate itemsets
Requires only 2 database scans	Multiple for each level of itemsets
Uses compact FP-Tree to compress data	Uses simple list-based representation
Faster and more scalable for large datasets	Slower due to expensive candidate generation
More memory efficient with compressed data	Uses more memory for candidates and support counts
Handles dense and long transactions better	Struggles with dense datasets and long frequent patterns

Apriori is a classical algorithm used for **frequent itemset mining** and **association rule learning** over transactional databases. It works on the principle that **all subsets of a frequent itemset must also be frequent** (Apriori property).

Working Steps of apriori:

1. **Scan the database** to find all frequent 1-itemsets.
2. **Generate candidate (k)-itemsets** from frequent (k-1)-itemsets using self-join.
3. **Prune** candidate itemsets whose subsets are not frequent.
4. **Scan the database** to count the support of candidates.
5. **Select frequent k-itemsets** with support $\geq$ minimum support.
6. **Repeat steps 2-5** until no more frequent itemsets are found.
7. **Generate association rules** from frequent itemsets with confidence $\geq$ minimum confidence.

Example: Consider a dataset with transactions:

Transaction ID	Items Bought
1	Bread, Milk
2	Bread, Diaper, Beer
3	Milk, Diaper, Beer
4	Bread, Milk, Diaper
5	Bread, Milk, Diaper, Beer

- Minimum Support = 60% (3 transactions)

Step 1: Find frequent 1-itemsets:

Item	Support Count
Bread	4
Milk	4
Diaper	4
Beer	3

Step 2: Generate candidate 2-itemsets: {Bread, Milk}, {Bread, Diaper}, {Bread, Beer}, {Milk, Diaper}, {Milk, Beer}, {Diaper, Beer}

Count support and select those ≥ 3 .

- {Bread, Milk} = 3
- {Bread, Diaper} = 3
- {Milk, Diaper} = 3
- {Diaper, Beer} = 3
- Others less than 3 $\rightarrow$ pruned

Step 3: Generate candidate 3-itemsets from frequent 2-itemsets: {Bread, Milk, Diaper}, {Milk, Diaper, Beer}, {Bread, Diaper, Beer}

Count support:

- {Bread, Milk, Diaper} = 2 (below 3, discard)
- {Milk, Diaper, Beer} = 2 (discard)
- {Bread, Diaper, Beer} = 2 (discard)

No frequent 3-itemsets.

Association rule mining discovers interesting relationships, patterns, or correlations between sets of items in large datasets. An **association rule** is typically expressed as: $A \Rightarrow B$.

Key Measures of Association Rules:

- **Support:** proportion of transactions containing both A and B.
 $\text{Support}(A \Rightarrow B) = \text{Count}(A \cup B) / \text{Total Transactions}$
- **Confidence:** likelihood that B appears containing A.
 $\text{Confidence}(A \Rightarrow B) = \text{Count}(A \cup B) / \text{Count}(A)$
- **Lift:** Measures the strength of the rule over random chance.
 $\text{Lift}(A \Rightarrow B) = \text{Confidence}(A \Rightarrow B) / \text{Support}(B)$

Applications of Association Rule Mining:

Application Area	Description
Market Basket Analysis	Understanding customer purchase behavior to identify product combinations frequently bought together.
Cross-Selling and Upselling	Suggesting related products to customers based on their purchase history.
Inventory Management	Optimizing stock based on frequent item combinations.
Web Usage Mining	Discovering patterns in web page visits to improve navigation and recommendations.
Fraud Detection	Identifying suspicious patterns in transactions.
Healthcare	Relate between symptoms and diseases for diagnosis, treatment planning.
Telecommunications	Detecting frequent call or message patterns for targeted marketing.

See visualisation of fp-tree, fp-growth, éclat, fp mining, candidate generation? Apriori